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## High Performance Computing

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| On Campus Exam                       | Closed Book                                                                                          |
|--------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Additional Materials:</b><br>None | <b>Permitted Materials:</b><br>Approved Non-Programmable<br>Scientific Calculators Permitted (Green) |

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**Guidelines:**

- The **duration** of this paper is 2 hrs
  - Word Count: No Word Count
  - Write all answers in the answer booklet provided.
  - Do not open the exam paper until instructed by the Invigilator.
  - Use only black pen or pencil.
  - Mobile and electronic devices are not allowed.
  - Do not communicate with other candidates during the exam.
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**Exam Instructions:**

Answer ALL Questions

### **Question 1**

- (a) The fastest computer on the June 2023 Top500 list was called Frontier. The following table shows information about the Frontier taken from the Top500 website.

|                                    |                                            |
|------------------------------------|--------------------------------------------|
| <b>Manufacturer:</b>               | HPE                                        |
| <b>Cores:</b>                      | 8,699,904                                  |
| <b>Processor:</b>                  | AMD Optimized 3rd Generation EPYC 64C 2GHz |
| <b>Interconnect:</b>               | Slingshot-11                               |
| <b>Linpack Performance (Rmax):</b> | 1,194.00 PFlop/s                           |
| <b>Theoretical Peak (Rpeak)</b>    | 1,679.82 PFlop/s                           |
| <b>HPCG performance:</b>           | 14,054.0 TFlop/s                           |
| <b>Power consumption:</b>          | 22,703.00 kW                               |
| <b>Operating system:</b>           | HPE Cray OS                                |

- (i) What is the LINPACK efficiency of Frontier? Show your working.

**(3 marks)**

- (ii) What is the theoretical peak performance of a single processor core?

**(5 marks)**

- (b) For each of the following loops(written in the C programming language) say whether it can be parallelised by adding OpenMP directives and still give the correct result, and explain why. You may assume that the loop gives correct results without OpenMP directives.

(i) 

```
for (int i=1; i<N; i++){  
    a[i]=0.2*a[i];  
}
```

**(4 marks)**

(ii) 

```
for (i=0; i<N-1; i++){  
    b[i]=0.25*(a[i+1]-a[i]);  
}
```

**(4 marks)**

(c) The Top500 list ranks the fastest supercomputers in the world according to the performance on the Linpack benchmark.

(i) What type of workload does the LINPACK benchmark use?

**(2 marks)**

(ii) “Given current trends soon all HPC systems are going to be commodity clusters.” Argue for **OR** against this position.

**(6 marks)**

(d) The following C program produces incorrect results when OpenMP is used:

```
#include <stdio.h>

const int N=100;

int main(){

    int sum=0;
#pragma omp parallel for default(none) shared(sum)
    for (int i=1; i<N+1; i++){
        sum+=i;
    }

    printf("Sum= %d\n", sum);
    return 0;

}
```

(i) Why are the results incorrect when OpenMP is used? You may assume that the results are correct when OpenMP is not used.

**(6 marks)**

(ii) Modify the program so that it produces correct results.

**(2 marks)**

(e) Effective load balancing can significantly improve the performance of a parallel program.

(i) Define what is meant by parallel efficiency?

**(2 marks)**

(ii) Explain what is meant by “load balancing”.

**(4 marks)**

(iii) How can load balancing be improved when parallelising a loop with OpenMP?

**(2 marks)**

**(Total 40 marks)**

## **Question 2**

- (a) The “Memory Intensive Service” is an HPC facility hosted by the University of Durham at the Institute for Computational Cosmology (ICC). The system comprises 452 compute nodes, each with 512 GB of RAM, and offering a total of 12 656 cores. All the compute nodes have the same number of processor cores.

- (i) What is the largest number of processor cores available to an OpenMP program running on this system and why?

**(2 marks)**

- (ii) What is the largest number of processor cores available to an MPI program running on this system and why?

**(2 marks)**

- (iii) Unified Parallel C can be used to write parallel programs as an alternative to OpenMP and MPI. According to Wikipedia:

Unified Parallel C (UPC) is an extension of the C programming language designed for high-performance computing on large-scale parallel machines, including those with a common global address space (SMP and NUMA) and those with distributed memory (e.g. clusters). The programmer is presented with a single shared, partitioned address space, where variables may be directly read and written by any processor, but each variable is physically associated with a single processor. UPC uses a single program, multiple data (SPMD) model of computation in which the amount of parallelism is fixed at program startup time, typically with a single thread of execution per processor.

Based on this information what is the largest number of processor cores available to a Unified Parallel C program running on this system and why?

**(2 marks)**

(b) You have been given the task of parallelising a program and need to achieve a strong scaling speed-up of at least 100.

(i) What is the smallest parallel fraction which permits a speed-up of 100?

**(3 marks)**

(ii) You parallelise the program and achieve a parallel fraction of  $p = 0.999$ . How many processors are required to obtain a speed up of 100 assuming idealised strong scaling?

**(3 marks)**

(iii) If the total parallel overhead is equivalent to 0.5% of the serial run time, what is the smallest number of processors required to achieve a speed up of at least 100?

**(4 marks)**



- (c) Consider a two-dimensional partial differential equation (PDE) solver which solves a PDE with one unknown variable  $u$ . To solve the PDE it is necessary to calculate values for the second derivative

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

This expression is evaluated using the finite difference stencil

$$\text{del2u} = (u(i+1, j) - 2*u(i, j) + u(i-1, j)) / (dx*dx) + (u(i, j+1) - 2*u(i, j) + u(i, j-1)) / (dy*dy)$$

The computation domain for the whole calculation comprises  $8192 \times 8192$  grid points. In addition to these grid points each sub-domain stores halo values which are copies of the data on neighbouring processors.

- (i) The calculation is domain decomposed in an  $8 \times 8$  configuration such that there are 64 square sub-domains of equal size. How many values need to be exchanged when one halo is sent to one neighbouring process?

**(2 marks)**

- (ii) How many bytes need to be sent if the values are represented using IEEE754 double precision?

**(2 marks)**

- (iii) What is the transmission time for sending a single halo over a network with a latency of  $1\mu\text{s}$  ( $10^{-6}\text{s}$ ) and a bandwidth of 12.5 GB/s (12.5 gigabytes per second)? Show your working.

**(3 marks)**

- (iv) The domain decomposition is modified so that the domain is decomposed in a  $16 \times 16$  configuration such that there are 256 square sub-domains of equal size. What is the transmission time for one halo with the modified domain decomposition?

**(2 marks)**

- (v) Modifying the domain decomposition has changed the ratio of communication time to computation time. By what factor has the ratio of communication time to computation time increased?

**(3 marks)**

- (vi) Why is the ratio of communication time to computation time *not* scaling as the number of grid points in the halo divided by the number of grid points in the sub-domain?

**(2 marks)**

**(Total 30 marks)**

### **Question 3**

- (a) You are given the task of planning an upgrade to an HPC cluster which involves replacing all the compute nodes. The total number of compute nodes will not change and there will be no other changes to the hardware. You need to choose between two compute node specifications:

- Compute node type 1: number of sockets=2, number of cores per socket=64, processor clock frequency=2.2 GHz, floating point operations per cycle =4
- Compute node type 2: number of sockets=2, number of cores per socket=28, processor clock frequency=2.8 GHz, floating point operations per cycle =16

The cost of the compute nodes is the same so you need to make a choice based on performance.

- (i) Which type of compute node has the higher theoretical peak performance? Show your working.

**(3 marks)**

- (ii) You know that the application which will run on the system has not been optimised to make effective use of the instruction set in modern processors. This means that the application is not able to use vector instructions and fused-multiply-add instructions. Explain whether this would influence your choice of compute node and why.

**(3 marks)**

- (iii) You know that the application which will run on the system makes extensive use of sparse matrix operations. What extra piece of information would you need to know in order to assess the suitability of the compute nodes for running this application and why is this information important?

**(2 marks)**

- (b) Show how the following sparse matrix would be represented in compressed sparse row format with rows and columns numbered starting from one.

**(10 marks)**

$$\begin{pmatrix} 7 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

- (c) A finite difference method is used to solve the diffusion equation. The time step used to compute the solution is calculated according to the constraint

$$\Delta t = \frac{1}{4} \frac{(\Delta x)^2}{K}$$

where  $\Delta t$  is the time step,  $\Delta x$  is the spacing between grid points, and  $K$  is a constant.

- (i) How does the computational cost of calculating a one-dimensional solution with  $N$  grid points compare with the computational cost of calculating a two-dimensional solution with  $N$  grid points in each dimension? You should assume that spatial resolution and number of time steps calculated are the same in both cases. Explain your reasoning.

**(2 marks)**

- (ii) How does the computational cost of the two-dimensional calculation increase if  $\Delta x$  is reduced by a factor of 10? You should assume that end time of the calculation and the extent of the computational domain remain constant. Explain your reasoning.

**(5 marks)**

- (iii) How does the time step constraint for the diffusion equation compare to the CFL condition for the advection equation and what effect does this have on the computational cost?

**(2 marks)**

- (iv) The diffusion equation can be solved using implicit time stepping which is unconditionally stable. What is meant by “unconditionally stable” and why can the time step not be arbitrarily large?

**(3 marks)**

**(Total 30 marks)**